



COLLATOR 200

Service Manual

Arte Sano LLC (Div. Fiesta Tortillas)
Calgary, AB

61391

Serial Number: 4588



Unless otherwise explicitly agreed in the written confirmation of order, for a period of twelve (12) months after delivery, the Seller warrants to the Purchaser:

1. That the materials and workmanship of the equipment are free from defects; and
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3. Any components or parts used in this equipment which are purchased from other manufacturers, in which case the original manufacturer's warranty shall apply;
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5. Damage to items of equipment caused by disasters such as fire, flood, wind and lightning;
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12. Unless modified by provisions contained in the confirmation of order, this Warranty is understood to be the complete and exclusive agreement between the parties, superseding all prior agreements, oral or written, and all other communications between the parties relating to the subject matter of this Warranty. No employee, other than an officer of the Company, or representative of the Seller is authorized to make any warranty in addition to those made herein.
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Manual Release & Revised Dates

Release/Revised Date	Version	Page(s)	Changes Made
June 2019	00001.0	110	Original Release
MM/YYYY	##	###	
MM/YYYY	##	###	
MM/YYYY	##	###	
MM/YYYY	##	###	
MM/YYYY	##	###	
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Contents

Introduction	7	Introduction
Safety	11	
Installation	25	
Setup and Changeover	29	
Operation	41	
Maintenance	59	
Troubleshooting	97	
Replacement Parts	103	
Index	107	

Customer Service and Support Group Outline

24-Hour Emergency Hotline

BluePrint Automation has incorporated a 24-hour hotline in the unlikely event you may need service.

Quality Trained Personnel

Personnel operating and servicing this equipment must be properly trained. **BluePrint** Automation strives to train our personnel to answer and deliver accurate solutions over the phone as well as in person. All service personnel are equipped with all the tools needed to fix as well as start up your equipment. They are also knowledgeable in assisting with all your spare parts needs. We have a team of service representatives located world-wide ready to assist you.

Response Time

BluePrint Automation is committed to respond to our customers in their times of need. Our commitment to every customer is to have service personnel there within 24 hours of your phone call if it is determined that on-site service is needed.

Spare Parts Group

BluePrint Automation has a professionally staffed spare parts group. This team will be able to help you with any parts needs along with kits that are specially designed for your equipment. Please feel free to call or email (orders@blueprintautomation.com) us for all your parts needs.

Need Help?

Call **001 804 520 5400** and when prompted select **Option 6**.

Your call will be answered by one of our representative. As soon as you hang up, representative will locate the best person from our team of qualified technicians to respond to your problem.

Contact Information



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About This Manual

Purpose

The purpose of this manual is to help you get the best value from your equipment. It provides information and procedures for operating, maintaining and servicing; and it offers diagnostic and repair procedures to follow when trouble occurs.

Using the manual

This manual:

- Introduction
- Safety
- Installation
- Setup and Changeover
- Operation
- Maintenance
- Troubleshooting
- Replacement Parts
- Index

Because this is an Operation and Service Manual it is not necessary for all personnel to read the entire manual cover to cover.

Operators

Operators should read the Introduction, System Overview, Safety, Operator information and Faults. Each of these sections will help familiarize them with safety procedures and Safety/Warning sign. Also, the operator will be introduced to the following:

- A brief description of system components
- Sequence of operation
- How to change recipe code parameters
- How to enter and change passwords
- And importantly, a description of the HMI

Maintenance Personnel

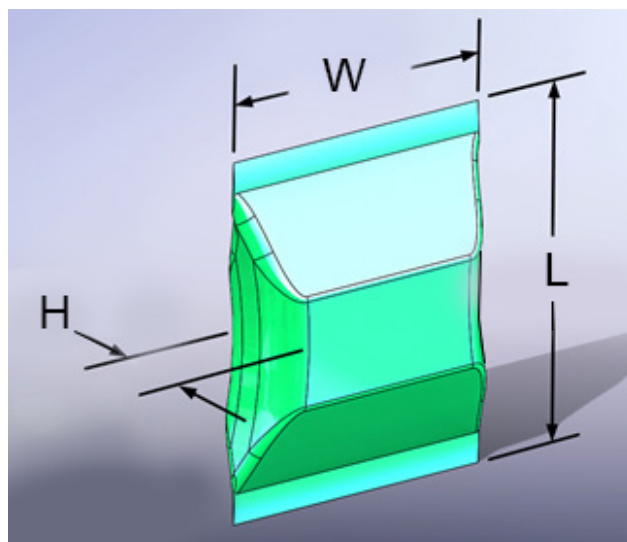
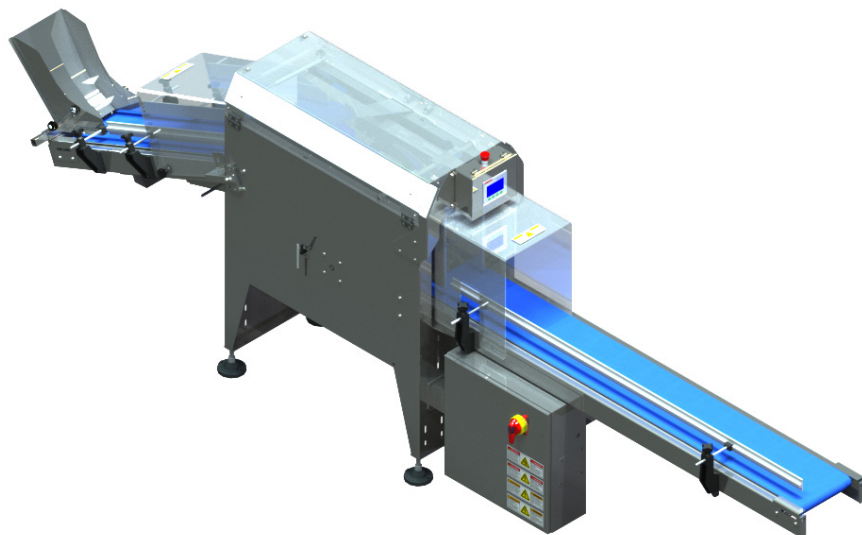
Maintenance personnel should read the entire manual to familiarize themselves with the safety precautions, safety devices, and safety signs used on this equipment. Maintenance will need to refer to each section at different times during maintenance.



Introduction

Description

The **Collator 200** is designed to receive packages from upstream equipment and reorient them such that they are arranged vertically collated front-to-back in a single row on the exit conveyor so that an operator can pick them up and place them into cases.



Standard Collator 200 Machine Limits:

Parameter	Units	Minimum	Maximum
Package Width (W)	in [mm]	4 [101,6]	12,5 [317,5]
Package Length (L)	in [mm]	4 [101,6]	17,25 [38,15]
Package Thickness (T)	in [mm]	0,75 [19,05]	5,25 [133,35]
Weight	lb [kg]	0,06 [0,027]	5 [2,268]
Rate	package/min	-	130
Cut off height of bagger	in [mm]	33,5 [850,9]	38 [965,2]

Glossary

The following lists and defines commonly used abbreviations, acronyms, and initialisms in the packaging machinery industry.

ARB – Activated Roller Belt conveyor: On the ARB™ Intralox conveyor, products rest on free-spinning angled rollers rather than on the belt surface. These rollers extend above and below the belt surface and are positioned at an angle in relation to the direction the belt travels. Rollers that are activated by the carryway surface below move products across the direction of roller orientation rather than the direction of driver travel, thus making it possible to move conveyed products selectively. ARB conveyor equipped machinery can change the direction, alignment, location, and speed of an item independently without using rails or complicated mechanical controls.

BCID – Bar Coded Identification Markings: A barcode is an optical, machine-readable representation of data; the data usually describes something about the object that carries the barcode. Traditional barcodes systematically represent data by varying the widths and spacings of parallel lines, and may be referred to as linear or one-dimensional.

CSSC – Center Special Slotted Case: The inner and outer flaps are different lengths. Both pairs of flaps meet at the center of the box. The inner flaps meet in so as to provide a level rest for the contents of the box.

DCC – Double Cover Case: Frequently used for tall or heavy products that would be difficult to lower into a box. The product is placed on the bottom cover, and the center is placed over the product. The two interchangeable covers (top and bottom) of the DC box are tray style and usually extend only a few inches down the center body.

EOAT – End of Arm Tooling: Or end effector is the device at the end of a robotic arm, designed to interact with the environment.

FLA – Full Load Amperage: the machine's current draw under full load in amperes (A).

FOL – The Full Overlap Case: The FOL configuration is ideal for boxes with relatively narrow widths. The outer flaps overlap the full width of the box making it especially resistant to rough handling. And, the extra thicknesses of corrugated material on the top and bottom panels provide added stacking strength.

HDP – Horizontal Delta Packer: A delta robot system that loads cases with product arranged horizontally in the case.

HMI – Human Machine Interface: Or System Control Interface is a touchscreen that provides the operator with an interface to start/stop the workcell and to select new products. Maintenance personnel use the screen for system maintenance, parameter setting, and troubleshooting. The HMI is located on the door of the Robot Control Enclosure.

HPP – Horizontal Pick and Place: Robotic gantry that loads cases with product arranged horizontally in the case.

HSC – Half-Slotted Box with Cover: The HSC is frequently used for both shipping and shelf storage, and in applications where the cover must be removed and replaced frequently. The top of the HSC box is a tray style which extends partially (usually less than 2/3 the depth of the bottom) PHSC, but can be made to extend the full depth of the bottom FHSC if extra strength is needed.

HVDP – Horizontal/Vertical Delta Packer: A delta robot system that loads cases with product arranged either horizontally or vertically in the case dependent on the operational setpoints and change parts installed.

HVPP – Horizontal/Vertical Pick and Place: Robotic gantry that loads cases with product arranged either horizontally or vertically in the case dependent on the operational setpoints and change parts installed.

ICE – Integrated Case Erector: Integrated robotic case erector.

KD – Knock Down: Corrugated cases fed to the integrated case erectors.

MCP – Main Control Panel: Enclosure that houses the system controls such as conveyor VFDs, PLC, power distribution devices, etc.

MGC – Modular Gate Case Packer: A case packer that loads the cases by dropping (gravity) the product into the case by means of a modular gate.

MPD – Missing Product Detector: A piece of machinery designed to weigh a case after the packing process in order to ascertain that the specified product quantity has been placed into the case; cases that are out of specification can then be rejected.

OSC – Overlap Slotted Case: style of box is similar to the RSC box, except the outer flaps overlap by an inch or more. The OSC style of box is used when the length of the box is considerably greater than the width and results in an excessively long distance between the end flaps. This makes it desirable to overlap the outer flaps to prevent them from spreading apart. The OSC box is usually closed with staples through the overlapping flaps.

PLC – Programmable Logic Controller: Or programmable controller is an industrial digital computer which has been ruggedized and adapted for the control of manufacturing processes, such as assembly lines, or robotic devices, or any activity that requires high reliability control and ease of programming and process fault diagnosis.

POL – Partial Overlap Case: With this box, all flaps have the same length. The outer flaps overlap by 1 inch or more.

PPE - Personal protective equipment: PPE refers to protective clothing, helmets, goggles, or other garments or equipment designed to protect the wearer's body from injury or infection. The hazards addressed by protective equipment include physical, electrical, heat, chemicals, biohazards, and airborne particulate matter.

RFID - Radio Frequency Identification: A wireless system comprised of two components: tags and readers. The reader is a device that has one or more antennæ that emit radio waves and receive signals back from the tag.

RGP – Rotary Gate Packer: A case packer that loads the cases by dropping (gravity) the product into the case by means of a rotary gate.

RSC – The Regular Slotted Case: is the most common box style used for shipping and storage. The two outer flaps of an RSC box meet at the center of the box when folded and are usually closed using tape, or staples.

RSD – Regular Slotted Display: This case is carefully designed, with the retailer in mind. Packing and closing is similar as an RSC case. There is no need to tear the front, as this case has a pre-cut front flap. This makes it easy to open, and shelf ready in mere seconds.

SCCR – Short Circuit Current Rating: a rating on components and assemblies representing the maximum level of short-circuit current that a component or assembly can withstand. This rating is used to determine compliance with NEC 110.10. It is measured in kilovolt-amperes (kVA).

SKU – Stock Keeping Unit: In the field of inventory management, a stock keeping unit is a distinct type of item for sale, such as a product or service, and all attributes associated with the item type that distinguish it from other item types. For a product, these attributes could include manufacturer, description, material, size, color, packaging, and warranty terms. When a business takes inventory of its stock, it counts the quantity it has of each SKU. SKU can also refer to a unique identifier or code that refers to the particular stock keeping unit.

VDP – Vertical Delta Packer: A delta robot system that loads cases with product arranged vertically in the case.

VFD – Variable Frequency Drive: A variable-frequency drive (also termed adjustable-frequency drive, variable-speed drive, AC drive, micro drive or inverter drive) is a type of adjustable-speed drive used in electromechanical drive systems to control AC motor speed and torque by varying motor input frequency and voltage.

VFFS – Vertical Form Fill and Seal machine: Vertical automatic packing machines using a single film coil to obtain a package with three seals: two transverse seals and one longitudinal seal. Its main field of application includes loose products, granulated products, doughy products or products difficult to handle. This type of machine is aimed at both food and non food markets.

VHPF – Vertical/Horizontal Pack Former: A case packing system that can load cases with product arranged either vertically or horizontally depending on installed change parts and operating parameter settings.

VPF – Vertical Pack Former: Case packing system that loads cases with product arranged vertically in the case.

VPP – Vertical Pick and Place: A robotic gantry system that loads cases with product arranged horizontally in the case.